

Comparative Analysis of Indoor Reconstruction Methods for Digital Twin Generation

Shintaro Ono*, Kerun Li*, Tomoki Kuzuhara*, Koji Kinoshita*, Senling Wang*, Hiroshi Takahashi*

* Graduate School of Science and Engineering, Ehime University, Japan

n807009c@mails.cs.ehime-u.ac.jp, n807044a@mails.cc.ehime-u.ac.jp, n807012x@mails.cc.ehime-u.ac.jp

kinoshita.koji.me@ehime-u.ac.jp, wang@cs.ehime-u.ac.jp, takahashi@cs.ehime-u.ac.jp

Abstract—This paper presents a comparative study of three indoor reconstruction methods from LiDAR point clouds for digital twin generation in robot simulation: BEV-only reconstruction, BEV + Heightmap reconstruction, and an O-Snap-like plane-detection-based baseline. Direct indoor reconstruction from raw point clouds remains challenging because point clouds are large, irregular, and sensitive to noise and density variation. To address this issue, we focus on bird’s-eye-view (BEV) representations, which provide a compact and computationally efficient description of room-scale indoor structures.

We evaluate the three methods on 40 rooms from S3DIS Area_2 using both quantitative and qualitative comparisons. The results show that BEV-only reconstruction provides the most consistent overall performance, while the plane-based baseline is effective in spaces dominated by simple planar floor-and-wall structures. The BEV + Heightmap method shows potential for representing localized height changes such as steps and stairs, although its performance is affected by remaining movable objects. These results clarify the strengths and limitations of BEV-based and plane-based design choices, and provide useful insights for selecting practical reconstruction strategies in digital-twin pipelines for robot simulation.

Index Terms—Indoor 3D reconstruction, bird’s-eye-view (BEV), heightmap, plane detection, digital twin, LiDAR point cloud

I. INTRODUCTION

Indoor 3D reconstruction from LiDAR point clouds is a fundamental problem in digital twin generation for robot simulation [1]. To build virtual environments that can support robot perception, navigation, and interaction experiments, it is essential to reconstruct the static structure of indoor spaces, especially room-scale elements such as floors, walls, and major boundaries [2]. In digital-twin-based robot development, such structural reconstruction serves as a basic layer for later simulation, validation, and Real-to-Sim environment construction [1], [3].

However, directly processing raw point clouds for indoor reconstruction remains challenging. Point clouds are often large, irregular, and sensitive to sensor noise and density variation, which makes direct modeling computationally expensive and sometimes unstable [4], [5]. Plane-based methods can represent major structures such as floors and walls effectively, but their performance tends to depend on the simplicity of the room geometry and the reliability of plane extraction [6]. In rooms with more complex layouts or local height variations,

such methods may fail to preserve important geometric characteristics [6].

To address these issues, this paper focuses on bird’s-eye-view (BEV) representations transformed from point clouds [2]. By projecting 3D point clouds onto the gravity direction and representing room occupancy in two dimensions, BEV provides a compact and computationally efficient representation of indoor structure [2]. This representation is particularly suitable for room-scale reconstruction, where the main objective is to capture clean floor-and-wall layouts rather than fine object details [2]. In addition, heightmap information derived on the same grid can supplement BEV by preserving local height variation, while plane-based modeling remains a useful baseline for comparison [2], [6].

Based on this motivation, this paper compares three indoor reconstruction methods:

- 1) BEV-only reconstruction, which models the room from horizontal occupancy and vertical extrusion;
- 2) BEV + Heightmap reconstruction, which incorporates local height variation into the generated mesh; and
- 3) an O-Snap-like plane-detection-based baseline, which reconstructs rooms using detected planar structures [6].

Rather than claiming a universally superior reconstruction algorithm, the goal of this study is to clarify the strengths and limitations of these design choices for digital-twin-oriented indoor modeling.

The main contribution of this paper is a comparative analysis of BEV-based and plane-based reconstruction strategies for indoor digital twin generation. Through quantitative and qualitative evaluation on indoor room data [7], we show that BEV-only reconstruction provides the most consistent overall performance, plane-based reconstruction is effective in simple floor-and-wall spaces, and heightmap-based reconstruction has potential for environments with localized height changes such as steps and stairs. These findings are useful for selecting practical reconstruction strategies in digital-twin pipelines for robot simulation.

The remainder of this paper is organized as follows. Section II describes the reconstruction methods and intermediate representations. Section III explains the evaluation methodology. Section IV presents the experimental results and discussion. Section V concludes the paper and outlines future work.

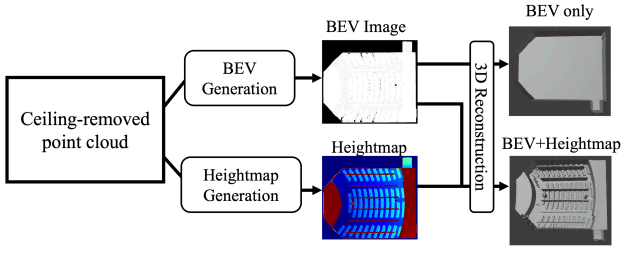


Fig. 1. **Overview of the reconstruction pipeline.** The input point cloud is converted into a BEV image and a heightmap representation. For visualization purposes, the heightmap is shown as an image in the figure, while in the actual implementation it is stored as a .npy file. Using these inputs, BEV-only and BEV+Heightmap reconstructions are generated.

II. METHODS

Fig. 1 shows an overview of the reconstruction pipeline. Given an input room point cloud, we first convert it into a BEV image and, when required, a heightmap defined on the same grid. Based on these intermediate representations, we compare three reconstruction methods.

A. BEV and Heightmap Generation

Given an input point cloud $P = \{(x_i, y_i, z_i)\}_{i=1}^N$, we first convert it into a bird’s-eye-view (BEV) image by projecting all points onto the horizontal x - y plane. The projected space is discretized into a 2D grid with a pixel size of 0.02 m. Each point is assigned to a grid cell according to its x - and y -coordinates, and cells containing one or more points are set to 255, while the remaining cells are set to 0. The resulting BEV image is denoted by $B(u, v)$, which is a binary occupancy image representing the horizontal room layout. Note that u and v denote image coordinate.

On the same grid, we also generate a heightmap to preserve local height variations. For each grid cell containing one or more projected points, the height value is defined as the maximum z -value of points in that cell relative to the minimum room height $z_{\min}$. The generated heightmap is denoted by $H(u, v)$. Empty cells are initialized with -1 . To reduce local gaps and small holes, morphological closing with a 9×9 kernel is applied. The heightmap used for reconstruction is stored as a .npy file; in Fig. 1, however, it is shown as a visualized color-mapped image for clarity.

B. BEV-only Reconstruction

The BEV-only method reconstructs the room structure using only the binary BEV image $B(u, v)$. Since the BEV image retains only horizontal occupancy information, this method approximates the room as a simple structure consisting of a single floor and vertically extruded walls.

First, the occupied region is extracted from the BEV image and treated as a 2D representation of the room layout. The contour of the occupied region is then used as the boundary of the room. To recover the vertical extent of the room, the minimum and maximum heights of the input point cloud, $z_{\min}$ and $z_{\max}$, are used. The floor is modeled as a single horizontal

plane located at $z_{\min}$, and the wall surfaces are generated by vertically extruding the 2D room contour from $z_{\min}$ to $z_{\max}$.

Because this method does not explicitly represent local height variation, it cannot model structures such as steps or staircases. However, its simple modeling procedure tends to produce reconstructions that preserve room-scale floor-and-wall structures well.

C. BEV + Heightmap Reconstruction

The BEV + Heightmap method reconstructs a height-aware surface mesh from the binary BEV image $B(u, v)$ and the heightmap $H(u, v)$. Unlike the BEV-only method, which approximates the room using a single floor plane and vertically extruded walls, this method directly generates a grid-based mesh whose local vertex heights are determined from the heightmap.

First, the BEV image is binarized and the occupied pixels are extracted. Each occupied pixel is then converted into one quadrilateral mesh element (quad). For each occupied pixel, four corner vertices are defined on the grid and shared with neighboring pixels when applicable. The x - and y -coordinates of these vertices are restored to the original metric coordinate system using the stored metadata, namely the minimum room coordinates, pixel size, and padding.

The height of each vertex is determined from the heightmap value at the corresponding grid location. Specifically, if the stored relative height is non-negative, the world-coordinate height is computed by adding the minimum room height $z_{\min}$; otherwise, $z_{\min}$ is used as a fallback value. By repeating this procedure for all occupied pixels, a quadrilateral surface mesh is constructed over the occupied region.

Because this method directly reflects local height variation in the generated mesh, it can represent non-flat structures more naturally than the BEV-only method. However, since the heightmap is computed directly from the input point cloud, residual movable objects such as desks and chairs may also appear as local protrusions in the reconstructed surface.

D. O-Snap-like Plane-Detection-Based Baseline

As a baseline for comparison, we implemented a plane-detection-based reconstruction method inspired by O-Snap [6]. The original O-Snap method combines automatic plane-based modeling with interactive refinement and optimization. In contrast, the present implementation focuses only on the automatic reconstruction part so that it can be fairly compared with the proposed automatic BEV-based methods.

Specifically, planes are iteratively extracted from the input point cloud using RANSAC. The detected planes are then classified into candidate floor planes and candidate wall planes according to their geometric orientation. From the detected floor candidates, the dominant horizontal plane is selected as the floor. For each wall candidate, the corresponding 3D points are projected onto a local 2D coordinate system defined on that plane, and the projected region is approximated by a rectangle. These rectangles are then transformed back into 3D space and converted into wall meshes.

Finally, the floor mesh and wall meshes are merged to obtain the reconstructed room model. Unlike the proposed methods, this baseline reconstructs room geometry directly from planar structures detected in the 3D point cloud, without converting the point cloud into a BEV representation.

III. EVALUATION

A. Dataset

We evaluated each method on 40 rooms included in Area_2 of the Stanford Large-Scale 3D Indoor Spaces Dataset (S3DIS) [7]. For each room, we prepared ground truth mesh data (GT) created from annotation data and reconstructed meshes generated by each method. We then quantitatively and qualitatively compared the GT with each method.

B. Evaluation Metrics

As quantitative evaluation metrics, we employed Chamfer Distance (CD) [8] in (1), Hausdorff Distance (HD) [8] in (2), and the F1-score in (5).

CD represents the average bidirectional distance error between point clouds sampled from the reconstructed model and the ground-truth model surfaces, HD represents the maximum bidirectional distance error between them, and the F1-score evaluates the balance between precision and recall of the reconstruction results. Lower values are better for CD and HD, while higher values are better for the F1-score.

$$CD(A, B) = \frac{1}{N_B} \sum_{j=1}^{N_B} \min_{a_i \in A} \|b_j - a_i\|_2 + \frac{1}{N_A} \sum_{i=1}^{N_A} \min_{b_j \in B} \|a_i - b_j\|_2 \quad (1)$$

$$HD(A, B) = \max \left(\max_{a \in A} \min_{b \in B} \|a - b\|_2, \max_{b \in B} \min_{a \in A} \|b - a\|_2 \right) \quad (2)$$

$$\text{Precision} = \frac{1}{N_A} \sum_{i=1}^{N_A} \mathbf{1} \left(\min_{b_j \in B} \|a_i - b_j\|_2 < \tau \right) \quad (3)$$

$$\text{Recall} = \frac{1}{N_B} \sum_{j=1}^{N_B} \mathbf{1} \left(\min_{a_i \in A} \|b_j - a_i\|_2 < \tau \right) \quad (4)$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

Here, A and B denote the point sets sampled from the reconstructed model and the ground-truth model, respectively, and N_A and N_B are the numbers of sampled points. In addition, τ denotes the distance threshold for F1-score computation, and $\mathbf{1}(\cdot)$ is an indicator function that returns 1 if the condition is satisfied and 0 otherwise.

Additionally, the generated models were visually inspected in Blender to qualitatively evaluate the reproduction of room shape and structural details.

TABLE I
QUANTITATIVE COMPARISON OF RECONSTRUCTION RESULTS

Method	CD↓	HD↓	F1↑
O-Snap-like	0.2131	1.1509	0.7447
BEV_only	0.1981	1.0437	0.7817
BEV+heightmap	0.2769	1.1301	0.6985

C. Evaluation Settings

The following two points describe the additional evaluation conditions.

- Since the 3D model generation process differs for each method, the number of vertices and surface subdivision density in the output meshes do not necessarily match. To mitigate the impact of this difference on evaluation results, this study introduced equal surface sampling. Specifically, evaluations were performed after uniformly sampling the same number of point cloud from each mesh surface.
- This study aims to reconstruct indoor structures useful for robot simulation, focusing primarily on floor and wall surfaces. Therefore, as a common preprocessing step for all methods, areas within 50 cm of each room's maximum height were removed, and point cloud excluding ceiling components were selected as the reconstruction targets.

IV. RESULTS AND DISCUSSION

A. Quantitative Comparison

TABLE I shows the quantitative comparison results for Area_2. Among the three methods, BEV-only demonstrated the best performance across all three evaluation metrics.

BEV-only achieved the lowest CD and HD, along with the highest F1-score, demonstrating that BEV representations enable stable overall reconstruction of indoor structures. Meanwhile, BEV + Heightmap showed the lowest average performance, while O-Snap-like yielded intermediate results.

B. Per-space Tendency

Comparisons across rooms also revealed that the effectiveness of each method depends on room characteristics. BEV-only achieved the highest F1-score in the most rooms, demonstrating high robustness across various indoor spaces.

The O-Snap-like baseline demonstrated good performance in spaces with relatively simple planar walls and floors, and successfully restored the main wall surfaces.

On the other hand, the heightmap-based method showed relatively good results in some corridor-like spaces. This suggests that utilizing height information may be effective in environments containing localized height variations, such as steps or stairs.

C. Qualitative Observation

Qualitative evaluation involving direct observation of completed 3D models provides additional insights that quantitative

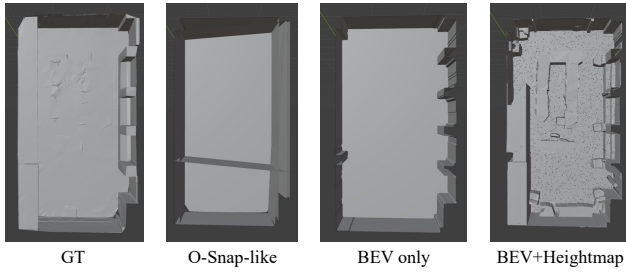


Fig. 2. **Qualitative comparison of reconstruction results for conferenceRoom_1.** BEV + Heightmap captures height-aware geometry, but also preserves unwanted objects such as desks and chairs. BEV only produces a simpler room structure closer to the target layout.

metrics alone cannot capture. In several cases, the BEV + Heightmap method visually reconstructed height-varying structures such as steps and staircases more naturally than other methods.

For example, Fig. 2 shows the reconstruction results for conferenceRoom_1 included in Area_2. In conferenceRoom_1, the heightmap-based reconstruction preserved floor height variations and the overall room geometry, but it also retained unwanted objects such as desks and chairs. Since these movable objects are not included in the ground-truth room structure, they likely degraded distance-based evaluation metrics such as CD and HD.

This observation suggests that the limitations of the Heightmap-based method may not necessarily lie in height representation itself, but rather stem from insufficient removal of movable objects prior to reconstruction. If object removal is improved, the Heightmap-based method has the potential to achieve performance beyond the current results.

D. Discussion

From these results, BEV-only demonstrated the most balanced results among the compared methods in terms of both structural reliability and reconstruction completeness. The O-Snap-like baseline was effective in spaces where walls and floors consisted of relatively simple planes, but it faced challenges in reproducing fine details in rooms containing complex structures. On the other hand, the BEV + Heightmap method, while inferior in average quantitative metrics, showed potential effectiveness in representing structures with height variations such as steps and stairs.

V. CONCLUSION

This paper compares three indoor reconstruction methods: BEV-only, BEV + Heightmap, and an O-Snap-like plane detection baseline. Quantitative evaluation on 40 rooms from S3DIS Area_2 revealed that BEV-only demonstrated the most consistent overall performance among the compared methods.

The O-Snap-like baseline proved effective in spaces with relatively simple planar structures. Meanwhile, although the BEV + Heightmap method performed worse on average quantitative metrics, it showed potential for representing structures with local height variations.

Future challenges include removing movable objects prior to heightmap reconstruction and developing a hybrid method combining the robustness of BEV representation with the advantages of plane detection-based modeling.

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